

EVIDENCE • CANVAS • SERIES

heatmap

A numeric matrix read as intensity — where the value concentrates across two dimensions.

Retention decays fastest in month two.

	M0	M1	M2	M3
Jan 2026	100	62	48	44
Feb 2026	100	58	44	41
Mar 2026	100	71	59	55
Apr 2026	100	69	57	

RETENTION · THE CEILING

Ten cohorts by twelve months — the density this holds.

	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11
January...	100	71	59	55	52	50	48	47	46	45	44	43
February...	100	58	44	41	39	38	37	36	35	34	33	32
March 2026	100	62	48	45	43	42	41	40	39	38	37	
April 2026	100	69	57	54	51	49	48	47	46	45		
May 2026	100	66	54	50	48	47	46	45	44			
June 2026	100	64	52	49	47	46	45	44				
July 2026	100	61	50	47	45	44	43					
August...	100	59	48	45	43	42						
September...	100	57	47	44	42							
October...	55	55	55	55								

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When NOT to reach for heatmap.

One row of numbers

A single series is not a matrix — it is a comparison, and a reader judges length far more precisely than intensity. Use `bar`. The kernel declines a flat list for this reason rather than painting a one-row grid.

Qualitative cells

If the cells are verbs, owners or statuses rather than numbers, the ramp has nothing to encode. Use `matrix-grid`, whose cells are tagged at parse time.

Precise comparison

Asking a reader which of two similar cells is larger spends the one thing intensity is bad at. If the comparison has to be exact, the value belongs on an axis — `bar` or `line`.

Geography

A value per country or region belongs on the `map` choropleth, which shares this ramp but places the cells where the reader expects them.

RELATED COMPONENTS

See also.

`matrix-grid` — the cells are qualitative — a verb, an owner, a status — rather than a number

`map` — the two dimensions are geographic; same ramp, spatial placement

`bar` — one dimension, and the comparison has to be exact

`quadrant` — two CONTINUOUS axes with items placed on them, rather than two categorical axes with a value at each crossing